AI CLUSTERS

AI-Ready Network Infrastructure: The Practical Realities of High-Power, Liquid-Cooled Rack Integration

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Agenda

- AI Fabrics at Scale
- Network as a Rack Shift
- Fiber and Optical Density
- Dealing with AI Density Challenges
- Call to Action

Networking Purpose Built for AI Clusters

As AI workloads demand bigger scale from compute, the network must keep up. The increase in network bandwidth has driven up the power and number of connections required at the switch level.

Switch ASIC Bandwidth

Legacy Switch
Up to 51.2 T

Switch ASIC power <500W
Air-cooled switch trays

AI Era Switch
102.4 T and beyond

Switch ASIC power >1000W
Liquid-cooled switch trays

Network Connections

Legacy Rack
100-200 fibers

TOR network switch integrated into rack with compute, storage, etc.

AI Era Rack
2000+ fibers

Networking deployed in separate switch racks

Power
The forcing function

Cooling
Air, Liquid, Immersion
Rack Scale Systems

Space
Not about “U” anymore

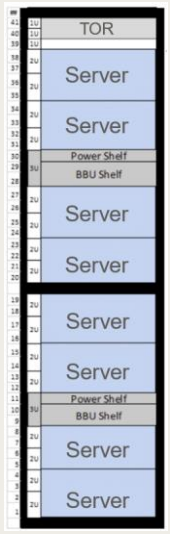
Efficiency and Flexibility

Networking is Now Deployed at Rack-Level

THE OLD MODEL

Individual switches
Air-cooled, procured and racked independently. Cabling, power, and cooling integration happen in the data center, one box at a time.

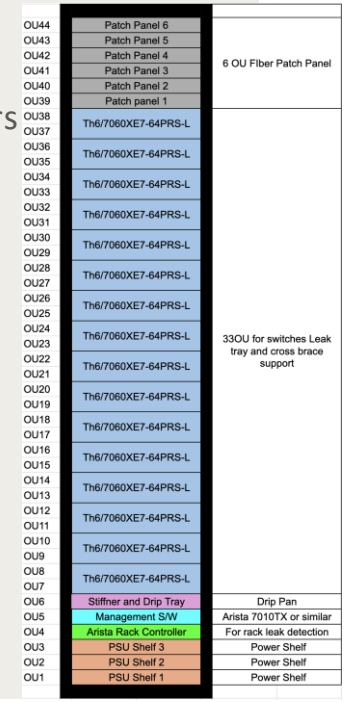
- Vendor ships switch chassis only
- Integration risk is local and contained
- Power: 1–2 kW per switch, standard PDU
- Fiber and optics managed per-device
- Cooling: room-level air handling



THE NEW MODEL

Integrated networking racks
Liquid-cooled; engineered, integrated, tested and delivered at the rack level.

- Vendor ships the full rack, wired and tested
- Integration risk is systemic — one rack, many vendors
- Power: 48V ORv3 busbar, 50 kW+ per rack
- Fiber and optics engineered at rack scale
- Cooling: direct-to-chip liquid loops



Sample Networking-Rack Elevations

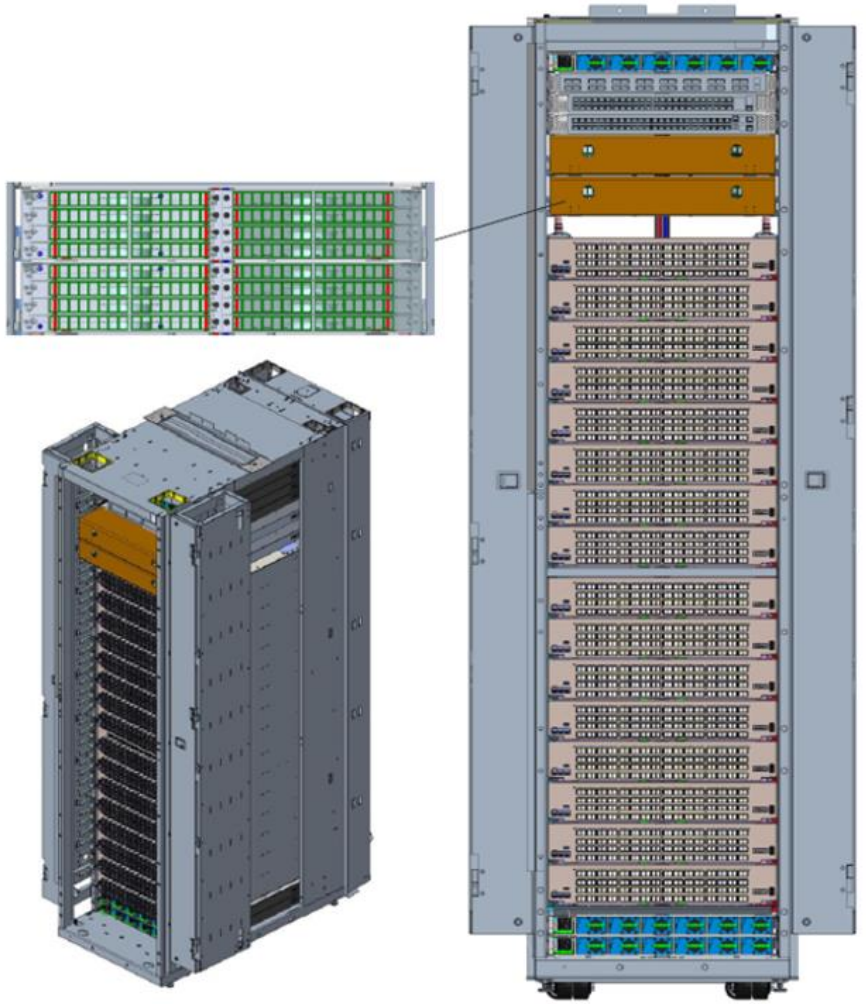
What changes in the AI-era networking rack

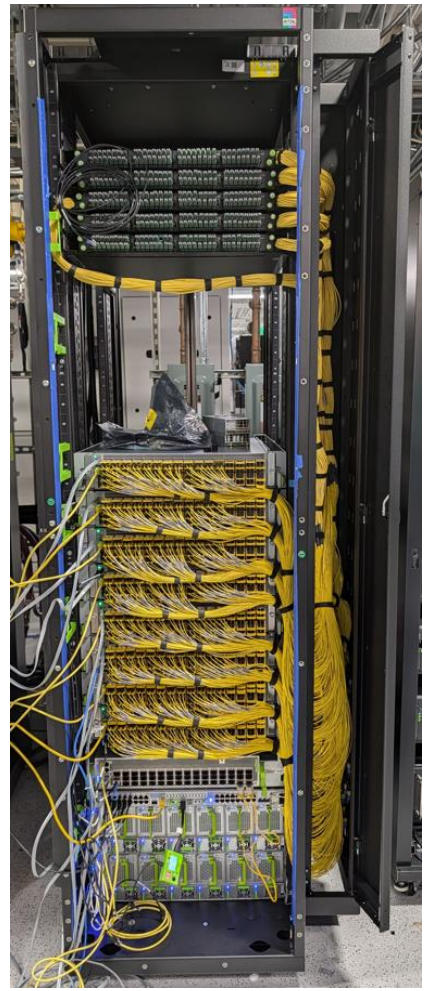
1. **Rear 48V ORv3 busbar**
   Blind-mate power connector on every switch tray; power shelves at the rack foot. Busbar geometry is fully prescribed by ORv3.

2. **Coolant manifold + quick disconnects**
   Vertical rear manifold feeds cold plates on each 51.2T tray; dripless flush-face QDs at every tray boundary.

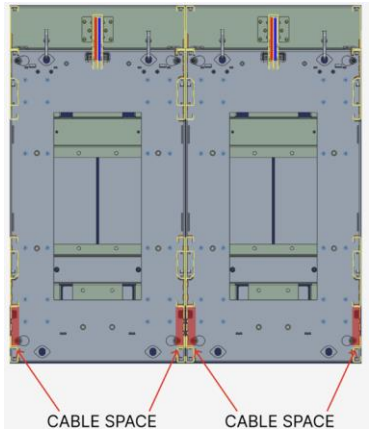
3. **Leak detection + rack controller**
   Loop-level sensing, flow/pressure telemetry and shut-off logic are baseline, not options.

4. **Vertical fiber management zones**
   Full-height routing for 2,000–4,000 fibers with protected bend radius, kept clear of the coolant loop.

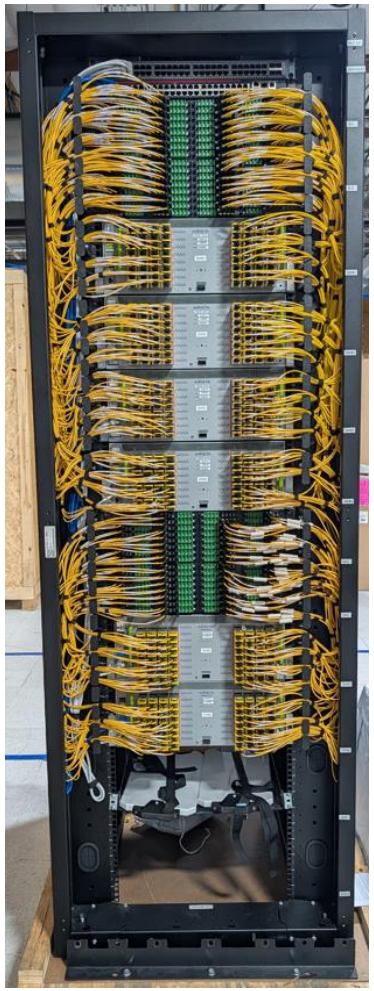


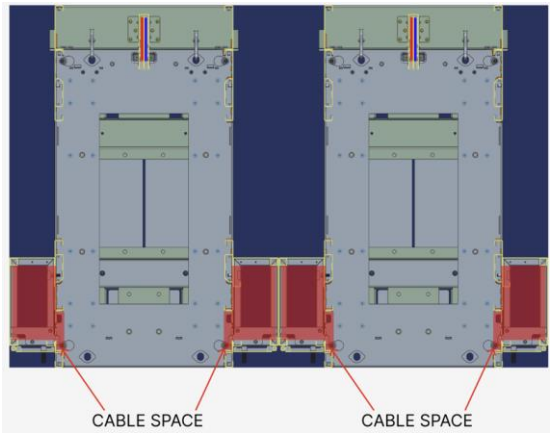


Cabling Challenges That AI Scale Brings

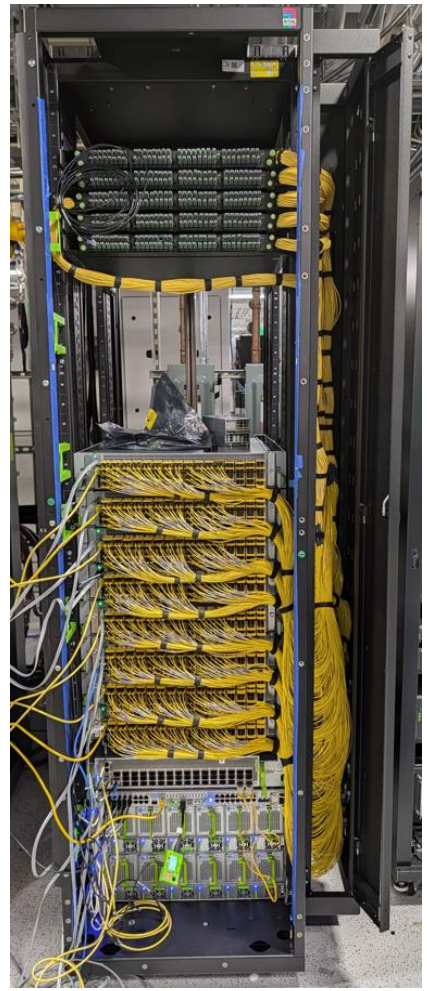


- Use of cable routing space available inside the rack
- Ideal cable routing but proved to be impractical
- Difficult to route cables without damaging them and/or violating bend radius spec





- Use of cable routing space on the outside of the rack
- Less than ideal for usage of floor space
- Much easier, and quicker, to assemble and cable this configuration reliably



Common Cooling Interfaces for AI Racks

OCP provides the building blocks. Networking racks need shared integration requirements to avoid incompatible designs.

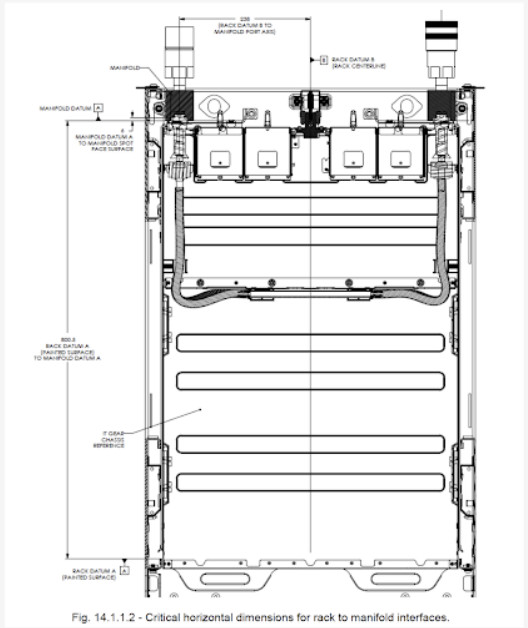


Fig. 14.1.1.2 - Critical horizontal dimensions for rack to manifold interfaces.

LIQUID COOLING (Port Location for switch mating interface)
-Defined in R-L and F-B directions
-Customer discrepancy lies in the Vertical OU space (Only referenced in the specification but leaves the positioning up to the customer switch)

Existing OCP foundation

- Published UQD, UQDB and ORv3 BMQC connector specifications
- Cold-plate and rack-manifold qualification guidance
- Coolant selection and wetted-material compatibility guidance

Networking rack gaps

- Manifold position and QD pitch across chassis heights
- Mating tolerances, clearances and service access
- Flow and pressure budgets for the complete cooling loop
- Leak detection, isolation and factory acceptance tests

Proposed OCP work

- Coordinate Cooling Environments, Rack & Power and Networking efforts
- Develop a shared rack interface profile and qualification matrix
- Validate interoperability across suppliers and share results

Which cooling interfaces can we standardize together?

Liquid Cooling Manifold Interface Details

- Manifold/chassis design is optimized for the 2U switch chassis
- Switch designs of chassis height other than 2U will require new manifold design
- Ideal (i.e. most flexible) manifold design would have QD at every OU location, requiring modified chassis design to eliminate QD to chassis interference





What Does Limited Commonality Cost?

Background info: The scale of AI is about more than just the number of racks and connections in a particular data center. The pace of data center construction, and the manufacturing of equipment to fill them, brings new challenges to the makers and customers in this space. Jabil and Arista alone has multiple customers requiring thousand of racks per year, 100s per week. Manufacturing efficiency is required to keep pace.

- Technical Impacts
  - The number of connections and cables in each rack make it challenging to produce AI-ready networking racks reliably, with high yields, at the volumes required.
    - New manufacturing test methods can help
      - AI-assisted visual inspection, intelligent network connectivity tests and other “new” techniques
- Supply Chain Impacts
  - The differences between customer ORv3 networking rack designs increases proprietary components, reducing leverage and driving up cost

Call to Action

- Ai-ready networking racks present many challenges around power, cooling, cabling, manufacturing and deployment
  - Commonality in the industry for cable management and thermal manifold/QD design in these networking racks will benefit makers and customers
- Join the related OCP Strategic Initiatives
  - https://www.opencompute.org/community/open-systems-for-ai
  - https://www.opencompute.org/community/open-data-centers-for-ai
  - https://www.opencompute.org/community/cluster-designs-for-ai
- Provide Feedback and reach out:
  - Arihant Jain (Arista) – ajain@arista.com
  - Rob Nance (Jabil) – rob_nance@jabil.com


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